

The Connection Center



THE UNIVERSITY OF
TENNESSEE
KNOXVILLE

Dylan Ballard, Lani Culley, Ethan Jones, Schumann Konen, Nick Spector, & Deason Swiatek

IE 404: Critical Design Review



THE UNIVERSITY OF
TENNESSEE
KNOXVILLE



Introduction

Client: Hope Hub Ministry within The Connection Center (powered by Cokesbury Church)

Sponsor: Kristine Inboden-Townsend, *Connection Center Director*

University Sponsor: Bill Hicks



Project Overview

What is The Connection Center?

- Day center serving unhoused guest in Knoxville
- Offers meals, showers, and essential resources
- Focus: **Hope Hub** → helping guests take next steps towards stability

The Problem

- No **standard** for decision-making
- **Inconsistent** guest experiences
- Volunteer-driven → varying **training & experience**
- Unreliable **data** & limited outcome tracking

The Ask

- Develop **SOPs** and **decision trees**
- Enable tracking of guest outcomes
- **Easy to use tools** for volunteers
- Improve **data accuracy & consistency**

Project Management

Project Management: Project Charter

Objective

Strengthen the Connection Center's ability to receive grants and funding through strong data. Allow the Connection Center to help more guests in the future.

Key Deliverables

Focused on one ministry to pilot and implement standardized processes, to create a scalable system that can be applied across other ministries utilizing Charity Check.

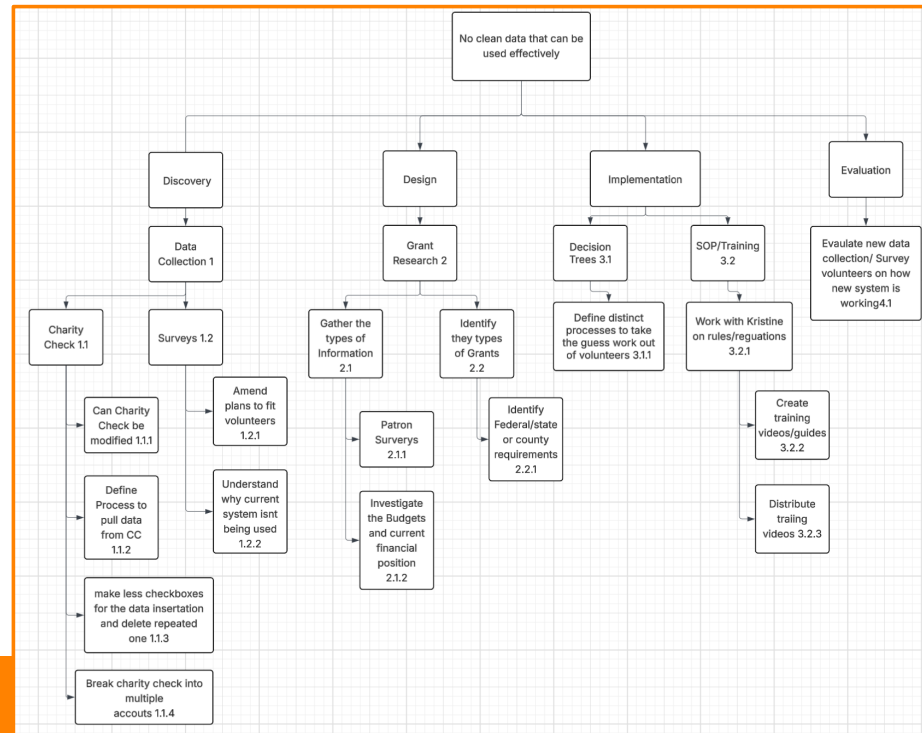
Risk & Constraints

Limited budget and reliance on volunteers require simple, intuitive solutions that do not depend on prior experience or complex training systems.

Communication Plan

Depends on volunteer adoption, active participation in surveys, and ongoing collaboration with staff and volunteers throughout the process.

Project Management: Work Breakdown Structure (WBS)



Project Management: Gantt Chart

Task	Responsible	Progress	Start Date	End Date	Days
Project Charter	Lani	Complete	2/1/26	2/20/26	-76
WBS	Nick	Complete	2/20/26	2/27/26	-69
Gantt Chart	Ethan	Complete	2/27/26	3/6/26	-62
Preliminary Design Review	Lani	Complete	3/6/26	3/20/26	-48
Midterm Presentation	Lani	Complete	3/20/26	3/27/26	-41
House of Quality	Nick	Complete	3/20/26	3/27/26	-41
Senior Design Checklist	Lani	Complete	3/27/26	4/3/26	-34
Poster Draft	Nick	Complete	4/3/26	4/10/26	-27
Critical Design Review Report Draft	Ethan	Complete	4/10/26	4/12/26	-25
Critical Design Review Presentation Draft	Lani	Complete	4/10/26	4/17/26	-20
Final Poster	Lani	Complete	4/17/26	4/24/26	-13
Writing Center Review	Deason	Complete	4/24/26	4/28/26	-9
Final Critical Design Review Report	Dylan	Complete	4/24/26	4/29/26	-8
Final Critical Design Review Presentation	Lani	Complete	4/30/26	5/4/26	-3
Website Development	Ethan	In Progress	4/30/26	8/15/26	107
Implementation - Decision Tree / Intake	Lani/Nick	In Progress	6/1/2026	8/15/2026	100
Evaluation	Dylan	Incomplete	10/6/2026	12/5/2026	212

Requirements Analysis (House of Quality)

Customer Requirements

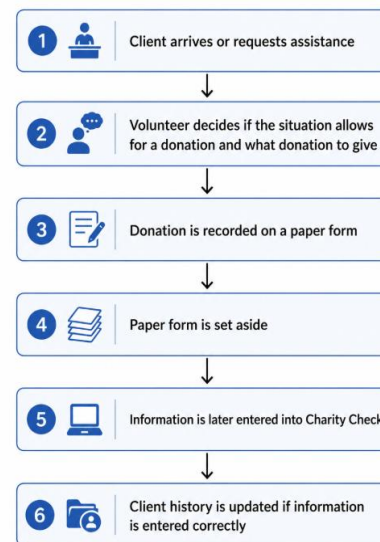
- Easy for volunteers to use with minimal training
- Accurate and consistent data collection
- Free to use
- Clear and structured decision-making

Technical Requirements

- Standardized digital intake form
- Decision tree for service allocation
- SOPs and volunteer training documentation
- User-friendly interface/layout

Current State and Proposed Solution

Baseline System (Original State)



Baseline System (Need for Improvement)

- Current process relies on paper forms – incomplete information
- Donation decisions are not standardized across volunteers
- Volunteers make most decisions under \$150 independently
- Inconsistent support and outcomes for guests
- Undefined workflow & ownership of processes
- Missing or incomplete client information impacts tracking
- Weak data reporting weakens grant and funding opportunities



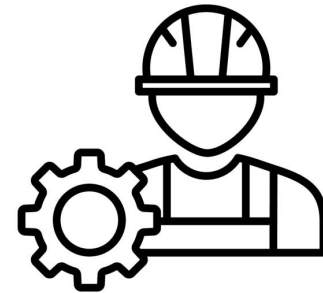
Design Considerations

- **Economic:** Developed a zero-budget solution
- **Sustainability & Reliability:** Created Standard Operating Procedures and decision trees
- **Ethical Standards:** Prioritized the protection of sensitive guest data
- **Social Impact:** Standardized the intake process to build guest trust



Industrial Engineering Methodology

- **Human Factors & Ergonomics:** reducing cognitive load
- **Lean Techniques:** Eliminate wasted steps
- **Statistical Analysis:** Analyzed inconsistencies
- **Industry Standards Alignment:** HMIS (Homeless Management Information System) and ISO 9001



Proposed Solutions - Website

**HOPE HUB**

Welcome Back

Sign in to access the document management system

Username

Password

[Sign In](#)

**HOPE HUB**

[Sign Out](#)

Document Categories

Select a category to view and manage documents

**SOPs**
1 document

Standard Operating Procedures and checklists

[View Documents >](#)

**Decision Trees**
0 documents

Flowcharts and decision-making guides

[View Documents >](#)

**Assistance Request Forms**
0 documents

Forms for requesting various types of assistance

[View Documents >](#)

Proposed Solutions – Intake Form

- Creating an online version
 - To be filled out by guest
 - Can be sent out beforehand to be filled out
- Eliminate unnecessary sections and focus on just getting the neighbor in front of a volunteer

Name (first last)		
Address		
City	State	Zip
Email		Phone
Birthdate	SSN	
Alternate Contact	Contact Phone	
Name of Spouse		
Number of Children in Household		
Number of Adults in Household		
Number of Seniors in Household		
Monthly Income \$:	Head of Household ☐ Yes ☐ No	
Gender	Male Female Other Unknown Refused	
Marital Status	Married Single Partnered Widowed Separated Unknown Refused	

Household / Utilities		
	Monthly	Balance
Mortgage or Rent	\$	\$
Electricity/Utilities	\$	\$
Water	\$	\$
Phone	\$	\$
Cable	\$	\$
Internet	\$	\$
Assets / Advantages		
Vehicle 1	\$	\$
Vehicle 2	\$	\$
Credit Card 1	\$	\$
Credit Card 2	\$	\$
Check Advance 1	\$	\$
Check Advance 2	\$	\$
Insurance / Medical		
Medical Insurance	\$	\$
Dental Insurance	\$	\$



Client Intake Checklist

This should be filled out by volunteers as they move through this checklist

* Indicates required question

Client Name *

Your answer

Date *

Date

mm/dd/yyyy

Intake Form *

☐ Signed Client Signature Page

☐ Verified SSN

☐ Make Copies of any vital d

New Client Intake Forum

Please fill this out directly from the client or from the paper copy

* Indicates required question

Name *

Your answer

Address (Address, City, State, Zip)

Your answer

Email

Your answer

Phone Number

Your answer

Birthdate

Your answer

Proposed Solutions – Decision Trees

The screenshot shows a web browser window with a single tab titled "Decision Tree - Requesting Assistance". The address bar displays the URL: https://docs.google.com/forms/d/1nDoopdlqXnZF2cwGaRlyg_nwmlc7lA/TnZCqsRXyFo/preview. The form is in "Preview mode" and includes a "Published" status indicator and a "Copy responder link" button. The form content features the "HOPE HUB" logo, which consists of a stylized network of nodes and lines with a heart in the center. Below the logo, the title "Decision Tree - Requesting Assistance" is displayed. A descriptive paragraph states: "This decision tree is designed to guide volunteers through a consistent and structured process when a guest requests assistance. It helps ensure fair, informed decisions by identifying the guest's immediate need, intended outcome, and next steps." At the bottom of the form, there are "Next" and "Clear form" buttons. A small disclaimer reads: "Never submit passwords through Google Forms." Below this, a note indicates the form was created inside of "Cokesbury Church" and asks if it looks suspicious, with a "Report" link. The "Google Forms" logo is at the very bottom of the form content area.

Testing Current Solutions – Summer 2026



Summer
Implementation:
Decision Trees &
Intake Form



Compliance
Checks Through
Data Exports



User Feedback
Survey's

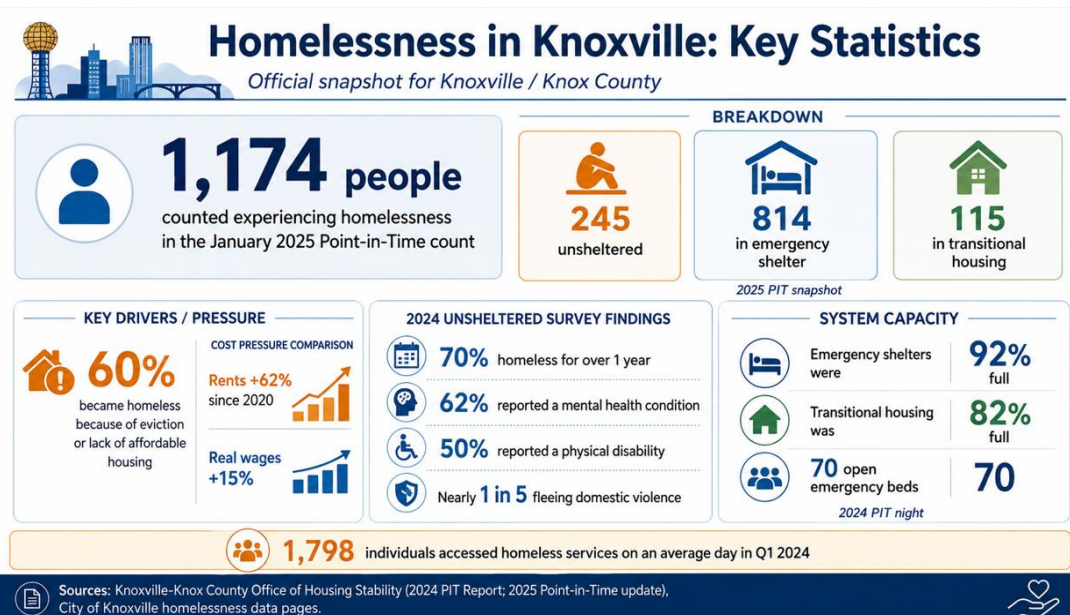


Fall: Evaluate
Summer Findings,
Begin SOP
Development, &
Rollout Website

Design Alternatives

Alternative	Strategy	Pros	Cons
1. High-Tech CRM	Implement automated software	High accuracy with real time reporting.	High cost; difficult for volunteers to learn.
2. Outsourced Management	Hire external data & grant consultants.	Professional results; no internal labor.	Expensive; not sustainable
3. Status Quo+	Monthly manual audits of existing files.	Zero cost; no change to current workflow.	Doesn't fix root cause.
Current Design	Standardized SOPs & Decision Trees	Low cost; volunteer friendly; scalable.	Requires initial culture shift in data entry.

Impact





Questions?



THE UNIVERSITY OF
TENNESSEE
KNOXVILLE